

COMPUTER SCIENCE — AI APPS

Offline AI Radio Medical Report Reviewer

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GitHub: github.com/Socram25Otsugua/AI-project---Medical-Report

The interface is divided into three main sections:

- Left Sidebar (Dark Blue):** Contains the 'Danish Maritime AUTHORITY - RADIO MEDICAL' logo, a 'New report' button with a plus icon, a search bar labeled 'Search reports...', a 'HISTORY' section with 'No saved reports yet.', and a user profile for 'Dr. M. Sørensen' (On-duty - Esbjerg).
- Top Center:** A structured workflow bar with eight steps: 1 P — Patient, 2 A — Airway, 3 B — Breathing, 4 C — Circulation, 5 D — Disability, 6 E — Exposure, 7 Problem, and 8 Actions. A status indicator shows 'Draft autosaved - just now'.
- Center Panel (Form):** Titled 'STEP 1 OF 8 Patient & vessel', it contains fields for patient and vessel information:
 - Patient & vessel:** Identity, vessel contact details, allergies and medicines.
 - Patient Fields:** FULL NAME (Maria), BIRTHDATE / CPR (2507974321), NATIONALITY (Portuguese), UTC TIME (04:18, 24h).
 - Vessel Fields:** SHIP NAME (MV Nordvind), SATELLITE CALL NO. (+870 123456789).
 - ADDITIONAL DETAILS:** GENDER (dropdown), DATE (YYYY-MM-DD), SHIPPING COMPANY, SHIP E-MAIL, CALL SIGNAL, and COORDINATES (55.6761° N, 12.5683° E).
- Right Panel (AI Doctor Chat):** Titled 'AI Doctor', it shows 'Online — responds in seconds'. The chat message reads: 'I'm your radio medical doctor for this session. Complete the form on the left, then send me the case when you're ready — I'll ask focused follow-up questions and help you reach a safe clinical summary.' There is a text input field 'Message the AI doctor...' and a send button.

Figure 0.1 — Radio Medical Assistant: Danish Maritime sidebar, eight-step ABCDE form, observation chart in the center panel, and dedicated AI doctor chat.

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1. Introduction and Problem Statement

1.1 Project overview

This project delivers an **offline AI assistant for maritime Radio Medical training**.

Participants complete a structured **Radio Medical Record (RMR)** form during simulation exercises. The system evaluates report quality, identifies clinical and documentation gaps, generates ABCDE-aligned treatment guidance, and supports a conversational “AI doctor” role-play aligned with Danish Radio Medical procedures.

The solution is deliberately **offline-first**: all inference runs locally via **Ollama**, retrieval uses a local **Chroma** vector store, and structured clinical signals are extracted through local **MCP tools** rather than cloud APIs. This matches training environments where connectivity is limited and patient data must remain on-premises.

1.2 Problem statement

Maritime medical officers must document patient assessments under pressure, often with incomplete information and limited shore-side support. Instructors need consistent, timely feedback on:

1. **Completeness** — Are ABCDE sections, vitals, problem description, and actions documented?
2. **Clinical safety** — Are abnormal vitals flagged and escalations considered?
3. **Actionable next steps** — What should the crew do now, monitor, and when to call Radio Medical again?
4. **Realistic telemedicine dialogue** — Can the trainee practice back-and-forth with an “AI doctor” before debrief?

Manual instructor review is time-consuming and varies between evaluators. Pre-written scenario responses lack adaptability when trainees omit key fields or document unexpected findings.

My goal: Build a reproducible, offline pipeline that ingests a trainee report, enriches it with protocol context (RAG + MCP), produces structured JSON feedback through LangChain chains, and exposes it via a React UI with session history — while keeping a human instructor in the loop for final judgment.

1.3 Frontend workflow (current design)

The React UI follows a single-session workflow inspired by Danish Maritime Radio Medical operations:

Area	Role
Sidebar	New report, search, saved history with clinical/completeness badges, on-duty profile
Stepper (8 steps)	P — Patient, A–E ABCDE, Problem, Actions
Center panel	Structured form; after first send to the AI doctor the form locks and the observation chart stays here
Right panel	AI doctor chat only — no observation chart embedded in the chat box
Header	Live session badge, progress tracker (step n / 8), View clinical summary button
Summary view	Clinical dashboard: patient hero banner, vitals grid, ABCDE recap, working impression

Trainees complete the ABCDE form and mandatory first-column observation vitals, send the case to the AI doctor, continue the dialogue in chat, add repeat observations in the center chart, and open the **clinical summary** when follow-up questions are resolved.

1.4 Architecture summary

Layer	Technology	Role
Frontend	React (Vite + TypeScript)	Stepper form, observation chart, chat, clinical summary, history
Backend	FastAPI	REST API, orchestration entry points
LLM	Ollama (llama3.1 default)	Form review, summary actions, patient eval, chat
Orchestration	LangChain	Prompt templates, JSON parsing, chains
RAG	Chroma + Ollama embeddings	Local guidelines and medicine chest
MCP	FastMCP + in-process tools	Vitals extraction, triage, checklist
Quality	Promptfoo + pytest	Prompt regression and unit tests

Backend services are grouped by responsibility:

- `app/services/form_analysis/` — form review chain (`analyze_form`)
- `app/services/chat/` — AI doctor turns and conversation state
- `app/services/summary/` — summary actions, patient evaluation, and `build_clinical_summary`

2. Model Choice, System Requirements, and Custom Model

2.1 Why Ollama and Llama 3.1

I selected **Ollama** as the inference runtime because it satisfies the offline requirement, supports structured JSON generation, and integrates cleanly with LangChain (ChatOllama, OllamaLLM, OllamaEmbeddings).

The default model is **llama3.1** with low temperature (0.1–0.2) for deterministic clinical outputs:

Chain	Model env var	Temperature	Rationale
Form review / Summary actions / Patient eval	OLLAMA_MODEL	0.2	Structured JSON; balance consistency and nuance
Chat doctor	OLLAMA_CHAT_MODEL	0.1	Conversational but stable tone

Why not a larger cloud model? Training deployments may lack reliable internet; data residency and latency favour local inference. Llama 3.1 offers sufficient instruction-following for JSON schemas at modest hardware cost.

Why not fine-tuning immediately? The domain knowledge is largely procedural (ABCDE, Danish maritime telemedicine). I inject that knowledge via **RAG** and **MCP** rather than weight updates, reducing data and compute requirements.

2.2 System requirements

Component	Minimum	Recommended
OS	macOS / Linux / Windows with WSL	macOS or Linux
Python	3.11+	3.11+
Node.js	18+	20+
Ollama	Latest, running locally	GPU-backed host for faster inference
RAM	8 GB	16 GB+ (LLM + Chroma)
Disk	~5 GB (model + vector store)	SSD preferred

Runtime services:

- Backend: `uvicorn app.main:app --reload --port 8000`
- Frontend: `npm run dev` → `http://localhost:5173`
- Ollama: `http://localhost:11434` with `llama3.1` pulled

2.3 Custom model: my approach and reasoning

I provide an **optional custom Ollama model** (`radio-medical-ai`) via a Modelfile that layers a domain system prompt and temperature on top of `llama3.1`. In production configuration I **default to the stock model** and rely on application-level system prompts, RAG, and MCP enrichment.

Arguments for a custom model

- Persistent domain persona without repeating long system prompts in every request.
- Slightly lower token usage per call.
- Easier hand-off to non-developer operators (`ollama create` once, set env var).

Arguments against (why I kept it optional)

- Prompt changes require rebuilding the model image rather than editing Python files.
- Promptfoo tests target `ollama:chat:llama3.1` directly; a custom tag adds another matrix dimension.
- Most behavioural control (JSON schemas, guardrails, vitals filtering) lives in **application code**, not base weights.
- For a training prototype, **prompt + RAG + MCP** delivered faster iteration than collecting fine-tuning data.

Conclusion: I document and ship the custom Modelfile for teams that want a packaged “Radio Medical AI” persona, but use stock `llama3.1` during development and Promptfoo CI unless profiling shows a measurable gain.

3. Prompts and Promptfoo Assertions

3.1 Prompt inventory

The backend defines four primary system prompts (see Appendix A for full text):

Prompt	File	Purpose
Form review	backend/app/prompts/form_review_prompt.py	ABCDE deficiency review, completeness score, safety flags
Summary actions	backend/app/prompts/summary_actions_prompt.py	Immediate actions, monitoring, escalation, follow-up questions
Patient evaluation	backend/app/prompts/summary_patient_eval_prompt.py	Patient status (ok / concerning / critical / unknown)
Chat doctor	backend/app/services/chat/chat_turn_service.py	Radio Medical Denmark role-play with case status

Each chain sends a JSON **user payload** containing `report_text`, `memory`, `rag_context`, and enriched `mcp` context. Outputs are parsed with LangChain `JsonOutputParser` bound to Pydantic schemas.

Design principles embedded in prompts:

- Do not invent missing vitals or history.
- Use MCP `field_presence` as hard evidence — do not mark documented fields as missing.
- Apply clinical threshold anchors (SpO₂, BP, temperature, consciousness).
- English-only output for instructor consistency.
- Chat doctor must not ask generic standing instructions already covered by an active conversation.

3.2 Promptfoo setup

Prompt contract tests live in `promptfoo/` (see `promptfoo/README.md`) and run in CI (`.github/workflows/pr-checks.yml`). I use the `ollama:chat:llama3.1` provider with `temperature: 0.2` and `format: json`.

Layout:

- prompts/ — review and response system prompts (aligned with backend/app/prompts/)
- fixtures/ — synthetic RMR training reports and stub review JSON
- assertions/ — reusable JavaScript schema checks
- promptfooconfig.review.yaml — review prompt regression
- promptfooconfig.response.yaml — response prompt regression

Run locally:

```
cd promptfoo
PLAYWRIGHT_SKIP_BROWSER_DOWNLOAD=1 npm install
export OLLAMA_BASE_URL="http://127.0.0.1:11434"
npm run test:prompts
```

3.3 Documented assertions

Review tests (promptfooconfig.review.yaml)

Test	Fixture	Assertions
Complete ABCDE report	fixtures/report_minimal.txt	contains-json; assertions/review-schema.cjs validates score bounds and array fields
Undocumented vitals	fixtures/report_missing_vitals.txt	contains-json; review- penalizes-incomplete- vitals.cjs asserts completeness_score <= 70

Response tests (promptfooconfig.response.yaml)

Test	Fixtures	Assertions
Action plan for complete report	report_minimal.txt + review_complete.json	contains-json; response-schema.cjs validates immediate_actions, monitoring, escalation, rationale, and questions arrays
Vitals gap for sparse report	report_missing_vitals.txt + review_sparse.json	contains-json; response-addresses- vitals-gap.cjs asserts follow-up questions or vitals-related immediate_actions

These assertions guard against model drift that returns non-JSON, drops required keys, or fails to penalize incomplete reports — without requiring exact string matches that would be brittle across stochastic LLM outputs.

4. Custom Ollama Model

Although optional, I ship a **Modelfile** at ollama/Modelfile:

```
FROM llama3.1

SYSTEM """
You are an offline clinical training assistant for maritime/remote medicine
simulations.
You must produce structured JSON outputs when asked.
Never invent patient data; explicitly mark missing fields.
"""

PARAMETER temperature 0.2
```

Create and use:

```
cd ollama
ollama create radio-medical-ai -f Modelfile
export OLLAMA_MODEL=radio-medical-ai
export OLLAMA_CHAT_MODEL=radio-medical-ai
```

The custom model does **not** replace application prompts — it sets a baseline persona. Form review, summary actions, and patient-eval chains still attach their full system prompts and JSON schemas. This layering lets operators swap the base tag while preserving chain-specific instructions.

When the custom model is **not** used, behaviour is equivalent because the same constraints appear in Python prompt constants and in Promptfoo prompt files under `promptfoo/prompts/`.

5. API Documentation

Base URL: `http://localhost:8000`

API prefix: `/api/v1` (configurable via `API_PREFIX`)

5.1 Health

Method	Path	Description
GET	<code>/health</code>	Liveness check → <code>{"ok": true}</code>

5.2 Report analysis

All report endpoints accept `ReportInput`:

```
{
  "session_id": "uuid-string",
  "report_text": "Full RMR text or form render",
  "locale": "en-UK"
}
```

Method	Path	Response	Description
POST	<code>/api/v1/reports/review</code>	<code>ReviewResult</code>	Deficiencies, safety flags, completeness and vitals scores
POST	<code>/api/v1/reports/respond</code>	<code>ResponseResult</code>	Treatment plan: actions, monitoring, escalation
POST	<code>/api/v1/reports/analyze</code>	<code>AnalyzeResult</code>	Full pipeline: review + summary actions + patient evaluation
POST	<code>/api/v1/reports/chat-turn</code>	<code>ChatTurnResult</code>	Conversational turn with AI doctor
POST	<code>/api/v1/reports/finalize-summary</code>	<code>AnalyzeResult</code>	Full analysis then clears session conversation state

Chat turn accepts `ChatTurnInput` (extends `ReportInput` with `user_message`).

Example — analyze request:

```
curl -X POST http://localhost:8000/api/v1/reports/analyze \
-H "Content-Type: application/json" \
-d '{"session_id":"demo-1","report_text":"Name: Test Patient\nSpO2: 93%\n..."}'
```

Key response shapes:

- ReviewResult — extracted, deficiencies[], safety_flags[], completeness_score, vitals_score, vitals_feedback
- ResponseResult — immediate_actions[], monitoring_parameters[], escalation_criteria[], rationale_bullets[], questions_for_participants[]
- PatientEvaluation — status, summary, suspected_problems[], red_flags[]
- ChatTurnResult — assistant_message, pending_questions[], can_finalize_summary

5.3 History (SQLite-backed)

Method	Path	Description
GET	/api/v1/reports/history?limit=50	List saved reports
POST	/api/v1/reports/history	Persist a report + result
DELETE	/api/v1/reports/history	Clear all history
DELETE	/api/v1/reports/history/{report_id}	Delete one entry

CORS is enabled for <http://localhost:5173> (Vite dev server).

6. Model Context Protocol (MCP)

I use MCP in two complementary ways: **deterministic FastMCP tools** for structured extraction, and **MCP-style context helpers** for RAG and session memory.

6.1 FastMCP server (backend/rmrr_mcp/medical_mcp_server.py)

Registered tools (invoked in-process via app/tools/mcp_client.py):

Tool	Input	Output
checklist_missing_sections	report_text	ABCDE section presence, field_presence map
extract_vitals	report_text	HR, SpO ₂ , RR, BP, temperature via regex
triage_priority	vitals	routine / urgent / critical + risk markers

Run standalone: `python -m rmrr_mcp.medical_mcp_server`

6.2 Context enrichment (app/mcp/)

`build_enriched_mcp_context()` merges:

- Vitals + triage + missing sections (from FastMCP tools)
- `guidelines_context` — RAG snippets from `medical_guidelines_mcp`
- `scenario_context` — detected scenario (trauma, cardiac, hypothermia, etc.)
- Session memory via `session_memory_mcp`

This bundle is injected into review, summary, and patient-eval chains so the LLM receives **evidence-backed** context rather than guessing completeness.

6.3 Why MCP alongside RAG?

Mechanism	Strength	Example
RAG	Semantic protocol retrieval	ABCDE escalation wording from guidelines
MCP tools	Deterministic parsing	Extract SpO ₂ : 93% → numeric 93
MCP checklist	Rule-based completeness	<code>field_presence.shipping_company == true</code>

Combining both reduces hallucinated “missing field” findings — a recurring failure mode when LLMs alone review structured forms.

7. Guardrails, LLM-as-a-Judge, and Human-in-the-Loop

7.1 Deterministic guardrails (`analysis_guardrails.py`)

After LLM output, I apply **rule-based filters**:

- **Contradiction filter** — Remove deficiencies/questions that claim a field is missing when regex/MCP shows it is present (e.g. shipping company, oxygen flow rate, pupil description).
- **Temperature filter** — Suppress hypothermia labels when documented temperature ≥ 35 °C.
- **Question deduplication** — Skip consciousness questions when AVPU/GCS already documented; filter generic “report any changes” chat questions.
- **Response normalization** — Ensure non-empty `immediate_actions`, `monitoring_parameters`, and `escalation_criteria` with safe defaults.

Vitals scoring (`vitals_coverage_score`) is computed entirely from MCP extraction — not from LLM judgment — giving instructors an objective 0–100 metric.

7.2 LLM-as-a-Judge

I considered a dedicated **judge chain** that re-scores or rejects primary outputs. The current design **embeds judgment into the review and patient-eval prompts** instead of a separate judge pass, to limit latency on local hardware.

Promptfoo JavaScript assertions act as a **lightweight automated judge** in CI: they verify schema compliance and business rules (e.g. sparse reports must yield follow-up questions). This is not a full LLM-as-judge stack, but catches regressions cheaply.

Trade-off: A separate judge model (e.g. “critic” pass) would improve safety at $\sim 2\times$ inference cost. For offline training, I prioritised deterministic MCP + post-filters.

7.3 Human-in-the-Loop (HITL)

The UI workflow enforces instructor oversight:

1. **Form** — Trainee enters data; nothing is final until reviewed.
2. **Chat** — AI doctor asks follow-ups; `can_finalize_summary` is false while `pending_questions` remain. The observation chart is edited in the center panel, not inside the chat box.
3. **Summary** — Full analyze runs when the trainee/instructor opens **View clinical summary**; results can be saved to history with clinical and completeness badges.

The system is advisory: it does **not** autonomously close cases or dispatch MEDEVAC. Chat prompts explicitly require Radio Medical sign-off formatting, reinforcing that outputs simulate training dialogue, not live clinical orders.

8. Conclusion

I delivered an offline Radio Medical training assistant that combines **LangChain orchestration**, **local RAG**, **MCP structured tools**, and a **React ABCDE workflow** with dedicated chat, observation chart, and clinical summary views. The architecture addresses the core problem: scalable, consistent feedback on trainee documentation and clinical reasoning without cloud dependency.

Key outcomes:

- Structured JSON API for form review, summary actions, patient status, and conversational turns
- Service modules split into form analysis, chat, and summary generation for clearer maintenance
- MCP-backed vitals extraction and completeness evidence reducing LLM contradictions
- Promptfoo regression tests wired into CI
- Optional custom Ollama model for operators who want a packaged persona
- Clinical summary dashboard with vitals quality dial, ABCDE recap, and working impression

For maritime simulation instructors, the system provides a practical copilot that accelerates debrief preparation while keeping humans accountable for final training judgments.

Appendix A — Prompt Reference

A.1 Form review system prompt (excerpt)

Source: `backend/app/prompts/form_review_prompt.py`

- Framework: ABCDE + MCP `field_presence` as hard evidence
- Output: JSON with `deficiencies`, `safety_flags`, `completeness_score`
- Clinical anchors: HR 60–80 normal; SpO₂ < 90% high risk; temp < 35 °C hypothermia

A.2 Summary actions prompt (excerpt)

Source: `backend/app/prompts/summary_actions_prompt.py`

- Outputs: `immediate_actions`, `monitoring_parameters`, `escalation_criteria`, `questions_for_participants`
- Prioritizes stabilization and escalation when safety flags present

A.3 Patient evaluation prompt (excerpt)

Source: `backend/app/prompts/summary_patient_eval_prompt.py`

- Status enum: `ok` | `concerning` | `critical` | `unknown`
- Must not invent missing vitals

A.4 Chat doctor prompt (excerpt)

Source: `backend/app/services/chat/chat_turn_service.py`

- Persona: Radio Medical Denmark
- Output JSON: `assistant_message`, `pending_questions`, `can_finalize_summary`

Appendix B — Promptfoo Configuration

B.1 Review config summary

File: promptfoo/promptfooconfig.review.yaml

```
providers:
  - id: ollama:chat:llama3.1
    config:
      temperature: 0.2
      passthrough:
        format: "json"
prompts:
  - prompts/review.txt
```

B.2 Response config summary

File: promptfoo/promptfooconfig.response.yaml

Same provider; prompt file prompts/response.txt.

B.3 Fixtures

File	Scenario
fixtures/report_minimal.txt	Structured ABCDE report with documented vitals
fixtures/report_missing_vitals.txt	Sparse report with undocumented vitals
fixtures/review_complete.json	Stub review for response tests (complete case)
fixtures/review_sparse.json	Stub review for response tests (incomplete vitals)

Appendix C — Environment Variables

Variable	Default	Description
APP_NAME	radio-medical-ai	FastAPI title
API_PREFIX	/api/v1	Router prefix
OLLAMA_BASE_URL	http://localhost:11434	Ollama host
OLLAMA_MODEL	llama3.1	Chain model
OLLAMA_CHAT_MODEL	llama3.1	Chat doctor model
OLLAMA_TEMPERATURE	0.2	Chain temperature
OLLAMA_CHAT_TEMPERATURE	0.1	Chat temperature
RAG_PERSIST_DIR	.chroma	Vector store path
RAG_COLLECTION	medical_training_kb	Chroma collection name

Appendix D — How to Export This Report to PDF

This report is **Markdown** and editable in Cursor, VS Code, or any text editor.

Option 1 — Project export script (recommended)

```
cd docs
node export-pdf.mjs
```

This writes docs/PROJECT_REPORT.pdf with the cover screenshot embedded.

To refresh the screenshot first (frontend must be running on port 5173):

```
cd docs
node capture-screenshot.mjs
node export-pdf.mjs
```

Option 2 — Cursor / VS Code

1. Install the **Markdown PDF** extension (yzane.markdown-pdf).
2. Open docs/PROJECT_REPORT.md.
3. Command Palette → **Markdown PDF: Export (pdf)**.

Option 3 — Browser print

1. View the rendered Markdown on GitHub or paste into StackEdit.
2. **Print** → **Save as PDF** with background graphics enabled.

Before submission checklist

- ☒ Course and student name filled in on the front page
- ☒ Screenshot reflects the current Radio Medical Assistant UI
- ☐ Export PDF and verify main body is **5–10 pages** (front matter and appendices excluded)
- ☐ Run `npm run test:prompts` in `promptfoo/` and attach results if required

End of report